TASK-CONDITIONED ENERGY-BASED WORLD MODELS FOR LATENT PREDICTIVE CONTROL

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ABSTRACT

The pursuit of general-purpose robotics requires agents capable of executing high-level natural language instructions in unstructured environments. Recent advancements in Model-Based Reinforcement Learning (MBRL) have demonstrated that planning within compressed latent manifolds, using techniques like latent Model Predictive Control (MPC), yields highly sample-efficient and robust continuous control. However, a fundamental limitation restricts the deployment of these optimization-based frameworks for language-guided tasks: standard latent MPC algorithms require an explicit representation of the target goal state at inference time to formulate the cost function, which is often infeasible in reality.

To bridge this gap, this thesis proposes a task-conditioned, energy-based latent predictive model that acts as an "imagination engine" for downstream control. By encoding visual observations via a Vision Transformer (ViT) and semantic task descriptions via a text embedding model (CLIP), the model autoregressively generates task-conditioned latent trajectories. A critical challenge in Joint-Embedding Predictive Architectures (JEPAs) is the risk of latent collapse. We address this by integrating Sketched Isotropic Gaussian Regularization (SIGReg), which provides theoretical guarantees against latent collapse. Additionally, we introduce a multi-step rollout loss to induce physical coherence in the model's "imagination."

Beyond control, establishing trust in autonomous robotic systems requires transparency in their internal decision-making processes. Because latent MPC operates in high-dimensional spaces, evaluating the safety of an intended action trajectory is inherently difficult. To address this, we train a convolutional decoder to project the model's autoregressive rollouts back into pixel space, effectively allowing human operators to interpret the robot's "thoughts" and verify its intent prior to physical execution. We evaluate the framework on a custom robotic manipulation dataset. Quantitative analysis utilizing a novel Task Sensitivity Index demonstrates the model's robust discriminative capability in grounding semantics. Qualitative geometric analyses via PCA, alongside decoded visual imaginations, reveal that the model synthesizes divergent, causally-accurate branches for counterfactual instructions. Ultimately, this work establishes a transparent representational foundation that provides the necessary predictive signals for future language-guided, trustworthy latent MPC frameworks.

LIST OF SYMBOLS OR ABBREVIATIONS

β	Task Sensitivity Index; ratio of counterfactual to concordant prediction error
γ	Parameters of the convolutional decoder
$\mathbf{e}_t$	Latent representation of o_t in $\mathbb{R}^d$
$\mathbf{z}$	Latent task representation in $\mathbb{R}^d$
$\mathcal{I}$	The manifold of all visual observations/images
Σ^*	The set of all language-based task descriptions
τ	Textual description of a specific task
$\tilde{o}$	Reconstructed visual observation
D_γ	Convolutional decoder for latent-to-image reconstruction
E_t, E'_t	Prediction error at time t for correct and counterfactual tasks, respectively
o_t	Visual observation at time t
W_ϕ	Learnable linear projection for task-vision latent alignment
z_{gain}	Scaling factor for the task latent to prevent task-marginalization
$\hat{\mathbf{e}}_{t+1}$	Predicted future latent state
λ_{RO}	Weighting hyperparameter for the rollout loss

λ_{SIGReg}	Weighting hyperparameter for the SIGReg loss
$\mathcal{N}(\mu, \sigma)$	Gaussian distribution with mean μ and variance σ^2
$\mathcal{S}^d$	The d-dimensional unit sphere
$a_{i:j}$	Sequence of quantity a from timestep i to j
g_ϕ	Neural encoder for task descriptions
g_θ	Neural encoder for visual observations
h_θ	Task-conditioned latent transition model
AMI	Autonomous Machine Intelligence
CLIP	Contrastive Language-Image Pre-training
EBM	Energy-Based Model
GCRL	Goal-Conditioned Reinforcement Learning
GPU	Graphics Processing Unit
HER	Hindsight Experience Replay
I-JEPA	Image Joint-Embedding Predictive Architecture
JEPA	Joint-Embedding Predictive Architecture
LoRA	Low-Rank Adaptation
MBRL	Model-Based Reinforcement Learning
MCMC	Markov Chain Monte Carlo
MPC	Model Predictive Control
PCA	Principal Component Analysis
RL	Reinforcement Learning
SIGReg	Sketched Isotropic Gaussian Regularization
Sim2Real	Simulation-to-Reality transfer

SSL	Self-Supervised Learning
SwiGLU	Swish-Gated Linear Unit
URLB	Unsupervised Reinforcement Learning Benchmark
UVFA	Universal Value Function Approximator
V-JEPA	Video Joint-Embedding Predictive Architecture
VAE	Variational Autoencoder
VICReg	Variance-Invariance-Covariance Regularization
VRAM	Video Random Access Memory

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